

Intrinsic Charge Transport in Lieb–Schultz–Mattis Gap Bounds

Qihang Wang,¹ Jinbo Wang,¹ and Kun Chen^{2,*}

¹*School of Mathematical Sciences, Peking University, Beijing 100871, China*

²*Institute of Theoretical Physics, Chinese Academy of Sciences, Beijing 100190, China*

(Dated: August 23, 2026)

Geometric interaction range can overestimate the energy cost of a Lieb–Schultz–Mattis twist. We derive finite-ring upper bounds on fractional-sector gaps from the twist phases connecting an interaction’s joint charge sectors. The shortest circular phase-arc radius is the exact minimum norm of a charge-algebra logarithm implementing the local twist, and its decomposition envelope is independent of a chosen interaction presentation. For a specified local term, a finite-dimensional transport quotient bounds this phase cost and has a closed wrapped-polarization formula for arbitrary-order correlated hopping. For a spin-1/2 chain we exactly evaluate the quadratic and finite-angle envelopes. On graph-fiber cylinders the bound is direction resolved: charge-conserving dynamics within a transverse slice has exactly zero longitudinal cost. Power-law moving tails give an explicit width–decay crossover. Even when every pair of transverse channels is coupled by a Euclidean dipolar tail, the bound vanishes if $W_L^2[1 + \log(L/W_L)] = o(L)$; a charge-moving van der Waals tail remains $O(W_L/L)$. For even longitudinal size, an odd-leg ladder corollary is independent of arbitrary translated $SU(2)$ -invariant rung dynamics. These are upper bounds on the gap, not evaluations of the physical gap itself.

The Lieb–Schultz–Mattis (LSM) obstruction turns fractional filling into an orthogonal low-energy state by inserting a slowly varying symmetry twist [1–5]. In a short-range chain, the resulting excitation energy is $O(L^{-1})$. Long-range quantum simulators make the underlying question sharper: should a coupling between distant sites pay according to its geometric range, or according to the charge it actually moves?

The distinction is physical. Exchange over a distance r acquires a twist phase of order r/L , whereas a density–density interaction is unchanged. Watanabe showed that twist-invariant density terms may be added without the decay restrictions needed for twist-sensitive hopping [6]. Ma treated k -local long-range interactions in arbitrary dimension and separated Type-I charge-moving tails from Type-II neutral couplings [7]; Zhou and Li obtained the one-dimensional two-body threshold [8]; and Liu *et al.* developed an anomalous-symmetry framework with long-range-entanglement consequences [9]. The Type-I and Type-II decay thresholds are recovered below only as consistency limits. The additional content here is a decomposition-independent Hamiltonian transport seminorm, its exact finite-angle phase-graph refinement, a wrapped-polarization rule for correlated hopping, exact XY seminorm values, and direction-resolved cylinder and dipolar-strip bounds.

Charge transport also appears in a state–unitary many-body index [10], while dipole-conserving models motivate correlated transfers [11]. Our object instead acts directly on a finite-volume Hamiltonian and bounds excitation energies without requiring global dipole conservation. A support-only substitution can assign exchange and density operators the same diameter cost; the weighted-charge commutant separates them and makes the optimized bound independent of a chosen interaction presentation

for the fixed lattice and on-site charge representation.

An effective one-dimensional Rydberg-dressed extended Bose–Hubbard model has now been realized experimentally [12]; related polar-molecule and Rydberg-dressed lattice models combine local kinetics with long-range interactions [13, 14]. Dipolar spin exchange and long-range XY dynamics have also been realized in molecular optical lattices and tweezer arrays [15–17]. For a periodic spin-1/2 chain, arbitrary translation-invariant diagonal tails—including dipolar or van der Waals interactions—drop out of the LSM energy estimate altogether. Their strengths may reshape the phase diagram, but they do not contribute to the uniform-twist cost.

Transport functional.— Let $\Lambda_L = \mathbb{Z}/L\mathbb{Z}$ and $\mathcal{H}_L = (\mathbb{C}^{2S+1})^{\otimes L}$. The integer on-site charge is $q_j = S_j^z + S\mathbf{1}$, with $Q_L = \sum_j q_j$. For a positive integer p , we assume only

$$[H_L, Q_L] = 0, \quad [H_L, T_L^p] = 0, \quad (1)$$

where $T_L q_j T_L^{-1} = q_{j+1}$. Eigenvalues are ordered with multiplicity, and $\text{gap}_{\text{full}}(H_L) = E_1 - E_0$. For an eigenvalue N of Q_L , write $E_{0,N} \leq E_{1,N} \leq \dots$ for the restricted spectrum on $\ker(Q_L - N\mathbf{1})$ and set $\text{gap}_N(H_L) = E_{1,N} - E_{0,N}$.

For a Hermitian B supported on X , the intrinsic finite-angle cost is read directly from the local twist. Let $\mathcal{P}_\lambda$ be the joint spectral projector of the commuting charges $\{q_j : j \in X\}$ at the integer tuple λ , and put $\theta_\lambda = (2\pi/L) \sum_{j \in X} j\lambda_j$ modulo 2π . Join λ and μ when $\mathcal{P}_\lambda B \mathcal{P}_\mu \neq 0$. For each connected component C of this canonical graph, let r_C be the radius of its shortest containing circular arc, and define

$$\eta_L(B) = \max_C r_C, \quad 0 \leq \eta_L(B) \leq \pi. \quad (2)$$

This definition is unchanged by basis rotations inside

degenerate charge eigenspaces. Moreover, $\eta_L(B)$ is exactly the minimum norm of a self-adjoint logarithm in the abelian algebra generated by the q_j whose conjugation on B equals the local twist (Supplemental Material).

When $[B, Q_X] = 0$, define

$$\mathcal{K}_X(B) = \left\{ c \in \mathbb{R}^X : \left[B, \sum_{j \in X} c_j S_j^z \right] = 0 \right\}. \quad (3)$$

This linear space records every weighted charge conserved by the term. With site representatives $j \in \{0, \dots, L-1\}$, its transport radius is

$$\tau_L(B) = \inf_{\substack{n \in \mathbb{Z}^X \\ c \in \mathcal{K}_X(B)}} \sum_{j \in X} |j + n_j L - c_j|. \quad (4)$$

The integer lifts make τ_L independent of representatives. A two-site exchange pays the cycle distance, whereas density interactions and polarization-neutral hopping can have arbitrary diameter but zero cost. For a specified finite-dimensional term, Eq. (4) is a concrete quotient optimization, and the rank-one case below is closed form. We do not claim that the subsequent infimum over all Hamiltonian decompositions is in general an efficient algorithmic problem.

The cost has a closed form for fixed-transfer processes. Let the nonzero product-basis charge-transfer rows of B span a one-dimensional space with primitive integer generator d , and let $x = (j)_{j \in X}$. Quotient-norm duality and Bézout's identity give

$$\tau_L(B) = \frac{\text{dist}(d \cdot x, L\mathbb{Z})}{\|d\|_\infty}. \quad (5)$$

For a spin-1/2 fixed-transfer monomial, $d_j \in \{-1, 0, 1\}$, so it costs zero exactly when its wrapped polarization displacement $\sum_j d_j j$ is a multiple of L . This classifies arbitrary-order correlated hopping, including processes that wrap the chosen seam.

The Hamiltonian quadratic transport moment is

$$\mathcal{T}_{2,L}(H_L) = \inf_{H_L = \sum_a B_a} \sum_a \tau_L(B_a)^2 \|B_a\|, \quad (6)$$

where the infimum is over finite Hermitian supported decompositions with $[B_a, Q_{X_a}] = 0$. The full-ring one-term decomposition is always available, and unions and rescalings of decompositions make $\mathcal{T}_{2,L}$ a seminorm on globally $U(1)$ -invariant Hermitian operators. The infimum is substantive: for the spin-1/2 XY ring, the bond decomposition attains it exactly, $\mathcal{T}_{2,L} = L|J|/2$, whereas the one-term presentation costs at least $|J|[L^2/4]^2$.

Its finite-angle companion is

$$\begin{aligned} s_{\text{ph},L}(B) &= \sin\left(\min\left\{\eta_L(B), \frac{\pi}{2}\right\}\right), \\ \mathcal{B}_{\text{ph},L}(H_L) &= \inf_{H_L = \sum_a B_a} \sum_a s_{\text{ph},L}(B_a)^2 \|B_a\|, \\ s_L(B) &= \sin\left(\min\left\{\frac{2\pi S}{L}\tau_L(B), \frac{\pi}{2}\right\}\right), \\ \mathcal{B}_{\text{sin},L}(H_L) &= \inf_{H_L = \sum_a B_a} \sum_a s_L(B_a)^2 \|B_a\|. \end{aligned} \quad (7)$$

All three decomposition infima are seminorms and canonical envelopes of their one-term costs (Supplemental Material). The logarithmic characterization and the quotient construction imply $\eta_L(B) \leq \min\{2\pi S\tau_L(B)/L, \pi\}$ and hence $\mathcal{B}_{\text{ph},L} \leq \mathcal{B}_{\text{sin},L}$. Equality $\eta_L(B) = 0$ holds exactly when B commutes with its local twist; this retains finite-angle cancellations that a real transfer-row space can lose.

Finite-ring bound.— For every eigenvalue N of Q_L with $pN \notin L\mathbb{Z}$,

$$\boxed{\begin{aligned} \text{gap}_N(H_L) &\leq 2\mathcal{B}_{\text{ph},L}(H_L) \leq 2\mathcal{B}_{\text{sin},L}(H_L), \\ 2\mathcal{B}_{\text{sin},L}(H_L) &\leq \frac{8\pi^2 S^2}{L^2} \mathcal{T}_{2,L}(H_L). \end{aligned}} \quad (8)$$

This statement requires neither a globally unique ground state nor prior knowledge of its charge sector. If the global ground state is unique in a fractional sector, $\text{gap}_{\text{full}}(H_L) \leq \text{gap}_N(H_L)$; for degenerate global ground energy the full gap is zero. Thus Eq. (8) contains the full-gap result as a corollary. The sector form is directly suited to canonical simulations and fixed-particle-number or fixed-magnetization preparations, without comparing ground energies between sectors. If H_L is globally $SU(2)$ invariant and $pS \notin \mathbb{Z}$, the corresponding full-gap bound holds: uniqueness would force a singlet, whose shifted charge produces the nontrivial unit-cell twist phase. A singlet is independently impossible when $LS \notin \mathbb{Z}$.

Long-range zero-cost interactions.— Specialize first to $p = 1$. Consider a spin-1/2 ring with $L \geq 6$ and real $J_\perp$,

$$H_L = J_\perp \sum_j (S_j^x S_{j+1}^x + S_j^y S_{j+1}^y) + D_L + Z_{\text{pol}}, \quad (9)$$

where D_L is any Hermitian, one-site-translation-invariant operator diagonal in the product S^z basis. It may contain arbitrary many-body diagonal terms. In particular,

$$D_L = \sum_j \sum_{r=1}^{\lfloor (L-1)/2 \rfloor} V_r S_j^z S_{j+r}^z - h \sum_j S_j^z \quad (10)$$

requires no decay condition on V_r for this finite-ring statement.

The second zero-cost class is explicitly off diagonal. Let Z_{pol} be any Hermitian, translation-invariant sum of fixed-transfer spin monomials, kept as separate terms, whose

primitive transfer vector d obeys $d \cdot x \in L\mathbb{Z}$. Equation (5) makes every such term zero cost. Its order, amplitude, and geometric span are unrestricted at fixed L . A four-body subfamily is, for $2 \leq r \leq \lfloor L/2 \rfloor$,

$$C_{j,r} = K_{r,L} S_j^+ S_{j+1}^- S_{j+r}^- S_{j+r+1}^+ + \overline{K}_{r,L} S_j^- S_{j+1}^+ S_{j+r}^+ S_{j+r+1}^-. \quad (11)$$

One charge moves across each of two separated bonds in opposite directions. With the lifted coordinates $(j, j+1, j+r, j+r+1)$, the transfer vector $(1, -1, -1, 1)$ has zero first moment. Hence $C_{j,r}$ commutes with that lifted polarization and $\tau_L(C_{j,r}) = 0$, including across the periodic seam. For even L at $r = L/2$, the translation orbit depends only on $\text{Re } K_{r,L}$; this is merely a redundant parametrization.

Every displayed constituent of $D_L + Z_{\text{pol}}$ has zero cost, although the atomic radius of their sum need not vanish. Keeping them separate proves that all three Hamiltonian seminorms of the sum vanish. Every non-trivial spin-1/2 zero-cost transfer annihilates the one-particle sector, whereas each transverse bond has norm $|J_\perp|/2$ and $\tau_L = 1$ when $J_\perp \neq 0$. One-particle matrix elements give the matching lower bound against every admissible decomposition (Supplemental Material), so $\mathcal{T}_{2,L}(H_L) = L|J_\perp|/2$ exactly. For $L \geq 6$, the phase witness also saturates the finite-angle polar constraints, giving $\mathcal{B}_{\text{sin},L}(H_L) = (L|J_\perp|/2) \sin^2(\pi/L)$. For every $L \geq 6$, the physical bond decomposition gives

$$\text{gap}_N(H_L) \leq L|J_\perp| \sin^2 \frac{\pi}{L} \leq \frac{\pi^2 |J_\perp|}{L}, \quad N \notin L\mathbb{Z}. \quad (12)$$

The first inequality is strictly sharper than the second for nonzero $J_\perp$. Both are independent of the strength, sign, range, and decay of the two zero-cost classes. If the global ground state is unique and lies in such a sector, the first right-hand side in Eq. (12) also bounds the full gap. The exactness proved above concerns the two Hamiltonian seminorms, not spectral saturation: Eq. (12) is an upper bound and need not equal the physical excitation gap.

The same mechanism covers a genuine unit cell and chiral hopping. Replace the transverse term in Eq. (9) by p -periodic real couplings

$$\sum_j [J_j (S_j^x S_{j+1}^x + S_j^y S_{j+1}^y) + D_j (S_j^x S_{j+1}^y - S_j^y S_{j+1}^x)]. \quad (13)$$

If $p \mid L$ and the zero-cost terms are also p -periodic, then every sector with $pN \notin L\mathbb{Z}$ obeys

$$\begin{aligned} \text{gap}_N(H_L) &\leq \frac{L}{p} \sin^2 \frac{\pi}{L} \sum_{a=0}^{p-1} \sqrt{J_a^2 + D_a^2} \\ &\leq \frac{\pi^2}{pL} \sum_{a=0}^{p-1} \sqrt{J_a^2 + D_a^2}. \end{aligned} \quad (14)$$

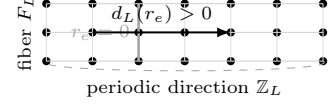


FIG. 1. Unrolled graph-fiber cylinder. A longitudinal twist charges a moving channel by its wrapped displacement $d_L(r_e)$; arbitrary charge-conserving intra-slice terms have $r_e = 0$.

Thus the explicit result includes uniform and dimerized XXZ-DM chains [18]. For $p = 2$ at half filling, $pN/L = 1$ and the hypothesis correctly fails, consistent with a symmetry-allowed dimerization gap.

Higher-dimensional flux-insertion arguments establish LSM obstructions on periodic systems [19]. The phase graph gives a directional dimensional-crossover consequence by resolving one periodic direction while assigning exactly zero longitudinal cost to transverse twist-invariant dynamics. The geometry is shown in Fig. 1. Let $\Lambda_L = \mathbb{Z}_L \times F_L$, $W_L = |F_L|$, and twist only the periodic coordinate, with finite-dimensional integer on-site charges. Assume $p \mid L$, $[H_L, Q_L] = [H_L, T_L^p] = 0$, and choose one oriented representative e of each physical Hermitian selective-hopping bond from (x, a) to $(x + r_e, b)$. If the edge collection and amplitudes are p -periodic, while $[D_L, Q_L] = [D_L, T_L^p] = 0$ and D_L has a supported decomposition into charge-conserving terms commuting with their local longitudinal twists, then every sector with $pN \notin L\mathbb{Z}$ obeys, with $d_L(r) = \min_{n \in \mathbb{Z}} |r + nL|$,

$$\text{gap}_N(H_L) \leq 2 \sum_e |t_e| \times \sin^2 \frac{\pi d_L(r_e)}{L}. \quad (15)$$

Transverse edges have $r_e = 0$ and cost exactly zero. For uniform fiber-preserving nearest-neighbor hopping, Eq. (15) is at most $2\pi^2 W_L |t|/L$; hence it vanishes for bounded hopping and $W_L = o(L)$. More generally, set $A_L(r) = \sum_{e: d_L(r_e)=r} |t_e|$. If $A_L(r) \leq CLK_L r^{-\alpha}$ with $\alpha > 2$, then

$$\text{gap}_N(H_L) \leq \frac{2\pi^2 CK_L}{L} \sum_{r=1}^{\lfloor L/2 \rfloor} r^{2-\alpha}. \quad (16)$$

Thus $K_L = O(L^\beta)$ gives a vanishing bound for $\beta < \min\{1, \alpha - 2\}$, with the logarithmic refinement at $\alpha = 3$. This condition concerns the actual longitudinal ℓ^1 moving-channel mass, not the degree of the transverse fiber.

For one-site translation ($p = 1$), there is a sharper geometric statement for dense channels. On $\mathbb{Z}_L \times \mathbb{Z}_{W_L}$, let every longitudinal moving amplitude obey $|t_e| \leq J(r_e^2 + s_e^2)^{-\alpha/2}$, where s_e is the transverse cycle distance and $W_L \leq \lfloor L/2 \rfloor$. Summing the Euclidean tail before applying

Eq. (15) gives, at dipolar decay $\alpha = 3$,

$$\boxed{\text{gap}_N(H_L) \leq \frac{2\pi^2 J W_L^2}{L} \left[3 + \log \frac{\lfloor L/2 \rfloor}{W_L} \right]}. \quad (17)$$

Hence fully connected dipolar exchange still forces a vanishing sector gap when $W_L^2[1 + \log(L/W_L)] = o(L)$; for a charge-moving van der Waals tail the same calculation is $O(W_L/L)$. This anisotropic aspect-ratio statement is not an isotropic two-dimensional LSM theorem. Its charge-conserving dipolar XY class is structurally relevant to molecular spin-exchange platforms [15–17]; we do not assert that those experiments realize the precise periodic strip geometry, filling, or spectral hypotheses used here.

The same formula is independent of unused local charge levels. For even L and an odd number of legs, a spin-1/2 SU(2)-invariant ladder gives $\text{gap}_{\text{full}}(H_L) \leq \pi^2 \sum_a |J_a|/L$, independently of arbitrary translated SU(2)-invariant intra-rung dynamics under the stated admissibility hypotheses. Standard fixed-width ladder LSM results are known [5, 20]; the point here is the explicit direction-resolved cost and its growing-width crossover.

For $V_r = V/r^\alpha$, a generic support-diameter substitution would add

$$\frac{\pi^2 |V|}{2L} \sum_{r=1}^{\lfloor (L-1)/2 \rfloor} r^{2-\alpha} \quad (18)$$

to the right side. At dipolar decay $\alpha = 3$, this support-only contribution is $O(\log L/L)$, while Eq. (12) is $L|J_\perp| \sin^2(\pi/L)$. At van der Waals decay $\alpha = 6$, both estimates scale as L^{-1} , and Eq. (12) remains independent of the diagonal tail. The polarization-neutral hopping terms go further: their order and geometric diameter can grow even though their uniform-twist phase cancels exactly. The transport budget therefore separates range from charge displacement for both diagonal and correlated-hopping interactions.

Proof mechanism.— Begin with any supported Hermitian decomposition $H_L = \sum_X \Phi_X$, without assuming termwise symmetry. Haar averaging,

$$B_X = \frac{1}{2\pi} \int_0^{2\pi} e^{i\theta Q_X} \Phi_X e^{-i\theta Q_X} d\theta, \quad (19)$$

preserves the sum, support, and Hermiticity, contracts the norm, and yields $[B_X, Q_X] = 0$. Thus global U(1), rather than termwise conservation, is the input.

Apply the uniform twist

$$U_L = \exp \left(\frac{2\pi i}{L} \sum_{j=0}^{L-1} j q_j \right). \quad (20)$$

The phase graph first replaces the local twist on B_X by an exact charge-algebra logarithm of minimum norm $\eta_L(B_X)$.

The transport quotient supplies the explicit logarithm

$$A_X(n, c) = \frac{2\pi}{L} \sum_{j \in X} (j + n_j L - c_j) S_j^z, \\ \|A_X(n, c)\| = \frac{2\pi S}{L} \sum_{j \in X} |j + n_j L - c_j|. \quad (21)$$

For self-adjoint A, B , paired twists cancel the linear response and obey the sharp finite-angle estimate

$$\left\| \frac{e^{-iA} B e^{iA} + e^{iA} B e^{-iA}}{2} - B \right\| \\ \leq 2 \sin^2 \left(\min \left\{ \|A\|, \frac{\pi}{2} \right\} \right) \|B\|. \quad (22)$$

Thus the phase cost is no larger than $s_L(B_X)^2$. The sharp paired inequality charges the term by $s_{\text{ph},L}(B_X)^2$; $\sin^2(\min\{x, \pi/2\}) \leq x^2$ then yields the full hierarchy in Eq. (8). Finally, U_L commutes with Q_L , while

$$T_L^p U_L T_L^{-p} = U_L e^{-2\pi i p Q_L / L}.$$

In a fixed sector $pN \notin L\mathbb{Z}$, a unique sector ground state and both of its twists have different T_L^p eigenvalues. The twists remain in the same sector, and one of their two excitation energies is bounded by the paired average. Sector min-max and the decomposition infimum give Eq. (8); a degenerate sector ground energy makes the sector gap zero. The appended Supplemental Material supplies the complete derivation, finite-range constants, phase-graph theorem, and transport certificates.

The same functional also recovers familiar results. Bounding τ_L by a support dispersion gives the standard $O(L^{-1})$ finite-range theorem with an explicit geometric constant. For power-law exchange it reproduces the one-dimensional Type-I vanishing threshold $\alpha > 2$, while separately neutral clusters reproduce the Type-II threshold $\alpha > 0$ [7]. Equation (8) retains the finite twist angle instead of relaxing it quadratically, while Eq. (12) evaluates the resulting budget for the physical model. More generally, any sequence with $\mathcal{T}_{2,L}(H_L) = o(L^2)$ has vanishing gaps in every chosen sequence of fractional charge sectors.

Certificates and geometric reduction.— The Supplemental Material gives primal and dual transfer-matrix certificates for τ_L , including correlated multi-charge transfers. Replacing the exact commutant by constant vectors recovers ordinary geometry. If every support has cycle diameter at most R , $L > 3R$, and $\sup_x \sum_{X \ni x} \|\Phi_X\| \leq J$, a median lift yields

$$\text{gap}_N(H_L) \leq \frac{8\pi^2 S^2 D_R^2 J}{L}, \quad D_R = \left\lfloor \frac{(R+1)^2}{4} \right\rfloor, \quad (23)$$

for every $pN \notin L\mathbb{Z}$. Equation (8) is never weaker and is strictly sharper when additional weighted charges are conserved.

In summary, spatial range alone is not the quantity seen by an LSM twist. The exact finite-angle term cost is the phase connectivity between joint charge sectors. For a specified term, the transport quotient explicitly certifies the displacement remaining after conserved weighted charges have been removed; the decomposition infimum then defines the Hamiltonian seminorm. Long-range diagonal forces may dominate the energy and determine the phase, yet they need not enter the finite-size LSM upper bound when they move no charge. These inequalities constrain, but do not calculate, the physical gap. On graph-fiber cylinders the same separation produces an explicit width-decay crossover, including the dense Euclidean dipolar-strip condition in Eq. (17), while assigning charge-conserving intra-slice dynamics zero longitudinal cost. This should be useful whenever local kinetics coexist with programmable long-range interactions.

K.C. acknowledges support from the Strategic Priority Research Program of the Chinese Academy of Sciences under Grant No. XDB1680102. Qihang Wang thanks his doctoral advisor, Weinan E, for his encouragement and support. We also acknowledge Gewu Intelligence Lab for providing the collaborative research environment in which this project was developed.

OpenAI GPT-5.6 Sol was used during proof exploration and drafting. Anthropic Claude Fable 5, operating in an agentic workflow, was used for adversarial checking, literature comparison, and language editing. The argument in this paper is self-contained, no model output is used as a mathematical premise, and the authors assume full responsibility for the final manuscript.

Data availability.— No data were created or analyzed. All results are analytic. Executable cross-checks were nonessential working aids; no research software is required to verify the proofs.

* chenkun@itp.ac.cn

- [1] E. Lieb, T. Schultz, and D. Mattis, Two soluble models of an antiferromagnetic chain, *Annals of Physics* **16**, 407 (1961).
- [2] I. Affleck and E. H. Lieb, A proof of part of Haldane's conjecture on spin chains, *Letters in Mathematical Physics* **12**, 57 (1986).
- [3] M. Oshikawa, M. Yamanaka, and I. Affleck, Magnetization plateaus in spin chains: Haldane gap for half-integer spins, *Physical Review Letters* **78**, 1984 (1997).
- [4] M. Yamanaka, M. Oshikawa, and I. Affleck, Nonperturbative approach to Luttinger's theorem in one dimension, *Physical Review Letters* **79**, 1110 (1997).
- [5] H. Tasaki, Lieb–Schultz–Mattis theorem with a local twist for general one-dimensional quantum systems, *Journal of Statistical Physics* **170**, 653 (2018).
- [6] H. Watanabe, A proof of the Bloch theorem for lattice models, *Journal of Statistical Physics* **177**, 717 (2019), arXiv:1904.02700.
- [7] R. Ma, Lieb–Schultz–Mattis theorem with long-range interactions, *Physical Review B* **110**, 104412 (2024), arXiv:2405.14949.
- [8] Y.-N. Zhou and X. Li, Validity of the Lieb–Schultz–Mattis theorem in long-range interacting systems (2024), arXiv:2406.08948 [cond-mat.str-el].
- [9] R. Liu, J. Yi, S. Zhou, and L. Zou, Entanglement area law and Lieb–Schultz–Mattis theorem in long-range interacting systems, and symmetry-enforced long-range entanglement, *Physical Review B* **112**, 214408 (2025), arXiv:2405.14929.
- [10] S. Bachmann, A. Bols, W. De Roeck, and M. Fraas, A many-body index for quantum charge transport, *Communications in Mathematical Physics* **375**, 1249 (2020), arXiv:1810.07351.
- [11] O. Dubinkin, J. May-Mann, and T. L. Hughes, Lieb–Schultz–Mattis-type theorems and other nonperturbative results for strongly correlated systems with conserved dipole moments, *Physical Review B* **103**, 125133 (2021), arXiv:2001.04477.
- [12] P. Weckesser, K. Srakaew, T. Blatz, D. Wei, D. Adler, S. Agrawal, A. Bohrdt, I. Bloch, and J. Zeiher, Realization of a Rydberg-dressed extended Bose–Hubbard model, *Science* **390**, 849 (2025), arXiv:2405.20128.
- [13] A. V. Gorshkov, S. R. Manmana, G. Chen, J. Ye, E. Demler, M. D. Lukin, and A. M. Rey, Tunable superfluidity and quantum magnetism with ultracold polar molecules, *Physical Review Letters* **107**, 115301 (2011).
- [14] M. Mattioli, M. Dalmonte, W. Lechner, and G. Pupillo, Cluster Luttinger liquids of Rydberg-dressed atoms in optical lattices, *Physical Review Letters* **111**, 165302 (2013).
- [15] B. Yan, S. A. Moses, B. Gadway, J. P. Covey, K. R. A. Hazzard, A. M. Rey, D. S. Jin, and J. Ye, Observation of dipolar spin-exchange interactions with lattice-confined polar molecules, *Nature* **501**, 521 (2013), arXiv:1305.5598.
- [16] L. Christakis, J. S. Rosenberg, R. Raj, S. Chi, A. Morningstar, D. A. Huse, Z. Z. Yan, and W. S. Bakr, Probing site-resolved correlations in a spin system of ultracold molecules, *Nature* **614**, 64 (2023), arXiv:2207.09328.
- [17] Y. Bao, S. S. Yu, L. Anderegg, E. Chae, W. Ketterle, K.-K. Ni, and J. M. Doyle, Dipolar spin-exchange and entanglement between molecules in an optical tweezer array, *Science* **382**, 1138 (2023), arXiv:2211.09780.
- [18] H. Karimi and I. Affleck, Transverse spectral functions and dzyaloshinskii–moriya interactions in XXZ spin chains, *Physical Review B* **84**, 174420 (2011).
- [19] M. B. Hastings, Lieb–Schultz–Mattis in higher dimensions, *Physical Review B* **69**, 104431 (2004).
- [20] P. Gagliardini, S. Haas, and T. M. Rice, Generalization of the Luttinger theorem for fermionic ladder systems, *Physical Review B* **58**, 9603 (1998), arXiv:cond-mat/9803035.
- [21] H. Bernien, S. Schwartz, A. Keesling, H. Levine, A. Omran, H. Pichler, S. Choi, A. S. Zibrov, M. Endres, M. Greiner, V. Vuletić, and M. D. Lukin, Probing many-body dynamics on a 51-atom quantum simulator, *Nature* **551**, 579 (2017), arXiv:1707.04344.

Supplemental Material for “Intrinsic Charge Transport in Lieb–Schultz–Mattis Gap Bounds”

Qihang Wang, Jinbo Wang, and Kun Chen

This Supplemental Material gives the complete fixed-sector proof, the joint-charge phase-graph theorem, universal trigonometric transport bounds, certificate formulas, the exact XY transport-seminorm evaluation, finite-range geometry, long-range corollaries, the wrapped-polarization selection rule, and the directional cylinder and spin-ladder bounds. It uses the notation and hypotheses of the main text.

PRECISE FINITE-RING STATEMENTS

Fix $L \geq 2$ and spin $S \in \{1/2, 1, 3/2, \dots\}$. All operator norms below are full Hilbert-space norms. A supported decomposition is finite; repeated or nonminimal support labels are allowed, although combining identical labels is harmless.

For an admissible supported term put

$$s_{\text{ph},L}(B) = \sin\left(\min\left\{\eta_L(B), \frac{\pi}{2}\right\}\right), \quad s_L(B) = \sin\left(\min\left\{\frac{2\pi S}{L}\tau_L(B), \frac{\pi}{2}\right\}\right) \quad (\text{S1})$$

and define $\mathcal{B}_{\text{ph},L}$ and $\mathcal{B}_{\text{sin},L}$ by Eq. (7).

Theorem A (decomposition-resolved sector bound). Let H_L satisfy Eq. (1). Consider any decomposition $H_L = \sum_X \Phi_X$ into Hermitian supported terms, and define B_X by Eq. (19). For every eigenvalue N of Q_L with $pN \notin L\mathbb{Z}$,

$$\text{gap}_N(H_L) \leq 2 \sum_X s_{\text{ph},L}(B_X)^2 \|B_X\| \leq 2 \sum_X s_L(B_X)^2 \|B_X\| \leq \frac{8\pi^2 S^2}{L^2} \sum_X \tau_L(B_X)^2 \|B_X\|. \quad (\text{S2})$$

No termwise symmetry or translation property is assumed of Φ_X .

Theorem B (Hamiltonian sector bound). Under the same hypotheses,

$$\text{gap}_N(H_L) \leq 2\mathcal{B}_{\text{ph},L}(H_L) \leq 2\mathcal{B}_{\text{sin},L}(H_L) \leq \frac{8\pi^2 S^2}{L^2} \mathcal{T}_{2,L}(H_L). \quad (\text{S3})$$

Corollary C (full-gap bound). If the global ground energy is degenerate, then $\text{gap}_{\text{full}}(H_L) = 0$. If the global ground state is unique and has charge $pN \notin L\mathbb{Z}$, then $\text{gap}_{\text{full}}(H_L) \leq \text{gap}_N(H_L)$, so Eq. (S3) also bounds the full gap.

Corollary D (half-odd-integer $SU(2)$). If H_L commutes with global $SU(2)$ and $pS \notin \mathbb{Z}$, the full-gap inequality in Corollary C holds. In particular, this includes half-odd-integer spin with one-site translation.

Every gap is defined with eigenvalues counted with multiplicity.

CONDITIONAL EXPECTATION AND TERMWISE SYMMETRY

Let $\mathbb{E}_X$ denote the $U(1)$ Haar average in Eq. (19). It is a unital completely positive projection onto the commutant of Q_X and is therefore contractive in operator norm. A direct finite-dimensional proof suffices here: every conjugate $e^{i\theta Q_X} \Phi_X e^{-i\theta Q_X}$ has norm $\|\Phi_X\|$, so the triangle inequality for the Bochner integral gives $\|\mathbb{E}_X(\Phi_X)\| \leq \|\Phi_X\|$.

Because Φ_X acts trivially outside X ,

$$e^{i\theta Q_X} \Phi_X e^{-i\theta Q_X} = e^{i\theta Q_L} \Phi_X e^{-i\theta Q_L}. \quad (\text{S4})$$

Linearity of the integral and global $U(1)$ invariance give

$$\begin{aligned} \sum_X B_X &= \frac{1}{2\pi} \int_0^{2\pi} e^{i\theta Q_L} H_L e^{-i\theta Q_L} d\theta \\ &= H_L. \end{aligned} \quad (\text{S5})$$

Hermiticity and support are preserved, and differentiation under the integral or direct Fourier decomposition shows $[B_X, Q_X] = 0$. Thus termwise charge conservation is a consequence of the global hypothesis, not an additional interaction contract.

LOCAL GAUGE REDUCTION

Let B be supported on X and commute with Q_X . Restricted to X , the global twist conjugation is generated by

$$G_X = \frac{2\pi}{L} \sum_{j \in X} j q_j. \quad (\text{S6})$$

For any $n \in \mathbb{Z}^X$, integer spectrum gives

$$e^{iG_X} = \exp \left[\frac{2\pi i}{L} \sum_{j \in X} (j + n_j L) q_j \right]. \quad (\text{S7})$$

Fix $c \in \mathcal{K}_X(B)$ and set $C = \sum_j c_j S_j^z$. All weighted on-site S^z charges commute with one another, and $[C, B] = 0$. Hence

$$e^{-iG_X} B e^{iG_X} = e^{-i(G_X - 2\pi C/L)} B e^{i(G_X - 2\pi C/L)}. \quad (\text{S8})$$

The difference between q_j and S_j^z contributes only a scalar to the generator and drops out of conjugation. The remaining generator is precisely $A_X(n, c)$ in Eq. (21).

For real coefficients a_j , the commuting operator $\sum_j a_j S_j^z$ has eigenvalues $\sum_j a_j m_j$ with independently selectable $m_j \in \{-S, -S+1, \dots, S\}$. Choosing each extremal sign gives

$$\left\| \sum_j a_j S_j^z \right\| = S \sum_j |a_j|. \quad (\text{S9})$$

This proves the norm identity in Eq. (21) as an equality. A minimizer in Eq. (4) is not needed: all subsequent estimates hold for every (n, c) and pass to an approximating sequence.

JOINT-CHARGE PHASE GRAPH AND EXACT LOGARITHM

We now prove the intrinsic finite-angle statement used in the main text. The argument also applies to arbitrary finite-dimensional sites with self-adjoint integer charges q_j , including degenerate charge eigenspaces. For every joint charge tuple $\lambda = (\lambda_j)_{j \in X}$, let

$$\mathcal{P}_\lambda = \prod_{j \in X} \mathbf{1}_{\{\lambda_j\}}(q_j), \quad \theta_\lambda = \frac{2\pi}{L} \sum_{j \in X} j \lambda_j \pmod{2\pi}. \quad (\text{S10})$$

The local twist is $V_X = \sum_\lambda e^{i\theta_\lambda} \mathcal{P}_\lambda$. The graph Γ_B has the nonzero projectors as vertices and an edge $\lambda \sim \mu$ whenever $\mathcal{P}_\lambda B \mathcal{P}_\mu \neq 0$. For a component C , set

$$r_C = \inf_{\alpha \in \mathbb{R}} \max_{\lambda \in C} d_{\mathbb{T}}(\theta_\lambda, \alpha), \quad \eta_L(B) = \max_C r_C, \quad (\text{S11})$$

where $d_{\mathbb{T}}$ is circular geodesic distance. Thus $2r_C$ is the length of a shortest closed arc containing the phases in C . Spectral projectors, rather than basis vectors in their ranges, make the definition invariant under charge-preserving basis rotations.

Let $\mathcal{A}_X^{\text{chg}}$ be the abelian algebra generated by the q_j . Its self-adjoint elements have the form $A = \sum_\lambda a_\lambda \mathcal{P}_\lambda$. Then

$$\eta_L(B) = \min \left\{ \|A\| : A = A^* \in \mathcal{A}_X^{\text{chg}}, e^{-iA} B e^{iA} = V_X^{-1} B V_X \right\}. \quad (\text{S12})$$

To prove the upper bound, choose for every component C a center α_C of a shortest containing arc and real lifts β_λ of θ_λ with $|\beta_\lambda - \alpha_C| \leq r_C$. Put $A = \sum_\lambda (\beta_\lambda - \alpha_C) \mathcal{P}_\lambda$. The unitary $D = \sum_\lambda e^{-i\alpha_C} \mathcal{P}_\lambda$ is constant on every graph component and hence commutes with B , while $e^{iA} = D V_X$. Therefore the conjugations agree and $\|A\| \leq \eta_L(B)$.

Conversely, if the conjugations agree, then $W = e^{iA} V_X^{-1}$ commutes with B . Since W is scalar on every joint charge subspace, each nonzero block $\mathcal{P}_\lambda B \mathcal{P}_\mu$ forces its two scalars to coincide. Thus W is constant on every component. The

coefficients a_λ consequently give real lifts of the component's twist phases after one circular shift, and $\max_{\lambda \in C} |a_\lambda| \geq r_C$. This proves the reverse inequality in Eq. (S12).

Applying Eq. (22) to the minimizing logarithm gives

$$\begin{aligned} & \left\| \frac{V_X^{-1} B V_X + V_X B V_X^{-1}}{2} - B \right\| \\ & \leq 2s_{\text{ph,L}}(B)^2 \|B\|. \end{aligned} \quad (\text{S13})$$

The reverse conjugation follows as well because A, V_X, W lie in the same abelian algebra and $[W, B] = 0$. The two-level example following Eq. (22) shows that the coefficient is sharp as a universal function of $\eta_L(B)$.

For the spin- S charge used in the main text, every lift and commuting weighted charge in Eq. (4) supplies an admissible charge-algebra logarithm through Eq. (21). Hence

$$\eta_L(B) \leq \min \left\{ \frac{2\pi S}{L} \tau_L(B), \pi \right\}, \quad s_{\text{ph,L}}(B) \leq s_L(B). \quad (\text{S14})$$

For general integer charges the same construction gives

$$\eta_L(B) \leq \frac{\pi}{L} \inf_{n,c} \sum_{j \in X} w_j |j + Ln_j - c_j|, \quad w_j = \max \text{spec}(q_j) - \min \text{spec}(q_j), \quad (\text{S15})$$

where $\sum_j c_j q_j$ commutes with B . Centering each q_j proves the norm bound in Eq. (S15).

The graph also gives the exact zero criterion

$$\boxed{\eta_L(B) = 0 \iff [B, V_X] = 0.} \quad (\text{S16})$$

Indeed, either condition says precisely that the twist phase is constant on every nonzero spectral-block component. For a spin-one-half fixed-transfer monomial with transfer vector r , the only nontrivial components are pairs, and

$$\eta_L(B) = \frac{\pi}{L} \text{dist} \left(\sum_j r_j j, L\mathbb{Z} \right). \quad (\text{S17})$$

Thus the phase graph agrees with wrapped polarization in this case but retains nonprimitive finite-angle cancellation at higher charge. For example, at $L = 6$ place two spin-one sites at 0 and 3 and take

$$B = | +1, -1 \rangle \langle -1, +1 | + | -1, +1 \rangle \langle +1, -1 |. \quad (\text{S18})$$

Its actual transfer $(2, -2)$ accumulates phase 2π , so $\eta_6(B) = 0$. The real-row transport radius instead uses the primitive row $(1, -1)$ and gives $\tau_6(B) = 3$. The intrinsic phase cost therefore detects an exact zero-cost process missed by that relaxation.

PAIRED-CONJUGATION INEQUALITY

Let $\mathcal{C}$ be the conjugation isometry $\mathcal{C}(B) = e^{-iA} B e^{iA}$. Algebraically,

$$\frac{\mathcal{C} + \mathcal{C}^{-1}}{2} - I = \frac{1}{2} \mathcal{C}^{-1} (\mathcal{C} - I)^2. \quad (\text{S19})$$

Write $a = \|A\|$ and $V = e^{-iA}$. If $a \leq \pi/2$, functional calculus gives

$$\|V - \cos(a)I\| \leq \sin a, \quad (\text{S20})$$

because $|e^{-it} - \cos a| \leq \sin a$ for $|t| \leq a$. Hence

$$\|(\mathcal{C} - I)(B)\| = \|[V, B]\| \leq 2 \sin(a) \|B\|. \quad (\text{S21})$$

For $a \geq \pi/2$, the trivial scalar choice zero instead gives $\|\mathcal{C} - I\| \leq 2$. Substitution in Eq. (S19) proves Eq. (22).

Combining Eq. (22) first with the exact logarithm and then with the transport upper certificate gives, term by term,

$$\begin{aligned}
& \left\| \frac{U_L^{-1} B_X U_L + U_L B_X U_L^{-1}}{2} - B_X \right\| \\
& \leq 2s_{\text{ph,L}}(B_X)^2 \|B_X\| \\
& \leq 2s_L(B_X)^2 \|B_X\| \\
& \leq \frac{8\pi^2 S^2}{L^2} \tau_L(B_X)^2 \|B_X\|.
\end{aligned} \tag{S22}$$

The finite-angle coefficient is sharp. Take $A_a = a \text{diag}(1, -1)$ and $B = \sigma_x$. Direct calculation gives

$$\left\| \frac{e^{-iA_a} B e^{iA_a} + e^{iA_a} B e^{-iA_a}}{2} - B \right\| = 1 - \cos(2a) = 2 \sin^2 a, \quad 0 \leq a \leq \frac{\pi}{2}. \tag{S23}$$

A two-dimensional block with this action can be placed between product basis states having equal total S^z , so local U(1) conservation alone does not improve the coefficient.

TRANSLATION PHASE, ORTHOGONALITY, AND MIN-MAX

Use representatives $0, \dots, L-1$ and $T_L q_j T_L^{-1} = q_{j+1}$. Reindexing Eq. (20) yields

$$\begin{aligned}
T_L U_L T_L^{-1} &= \exp \left(\frac{2\pi i}{L} \sum_{j=0}^{L-1} (j-1) q_j \right) e^{2\pi i q_0} \\
&= U_L e^{-2\pi i Q_L / L},
\end{aligned} \tag{S24}$$

where $e^{2\pi i q_0} = \mathbf{1}$. Iterating this identity gives

$$T_L^p U_L T_L^{-p} = U_L e^{-2\pi i p Q_L / L}. \tag{S25}$$

Both H_L and T_L^p preserve each charge sector, and U_L commutes with Q_L . Fix an eigenvalue $pN \notin L\mathbb{Z}$. If the sector ground state $\Psi_{L,N}$ is unique, it is a T_L^p eigenvector. The phase in Eq. (S25) changes that eigenvalue for either twist sign, so $U_L^{\pm 1} \Psi_{L,N}$ remain in the same sector and are exactly orthogonal to $\Psi_{L,N}$.

Let

$$e_\sigma = \langle U_L^\sigma \Psi_{L,N}, H_L U_L^\sigma \Psi_{L,N} \rangle - E_{0,N}, \quad \sigma = \pm 1. \tag{S26}$$

Both values are nonnegative. Summing Eq. (S22) and taking a ground-state expectation gives

$$\frac{e_+ + e_-}{2} \leq 2 \sum_X s_{\text{ph,L}}(B_X)^2 \|B_X\|. \tag{S27}$$

One sign lies below the right-hand side and is an orthogonal normalized trial state in the same sector. The finite-dimensional sector min-max principle proves Eq. (S2). If the sector ground energy is degenerate, the same inequality is automatic because $E_{1,N} = E_{0,N}$. A fractional sector has dimension at least two: the only one-dimensional charge sectors have every site in the lowest or highest charge state, for which $pN/L \in \mathbb{Z}$.

For the SU(2) full-gap corollary, uniqueness forces a spin singlet, so $N_L = LS$. Its cell-translation phase is nontrivial exactly when $pS \notin \mathbb{Z}$. If $LS \notin \mathbb{Z}$, a singlet is impossible because Q_L has integer spectrum.

HAMILTONIAN TRANSPORT SEMINORMS

Equation (S2) holds for every admissible decomposition, and the full-ring decomposition with one term H_L is always admissible. Taking the corresponding decomposition infima proves Eq. (S3).

If $H = \sum_a B_a$ and $K = \sum_b C_b$ are admissible decompositions, their union is an admissible decomposition of $H + K$. Therefore all three functionals are subadditive; in particular,

$$\begin{aligned}\mathcal{B}_{\text{ph},L}(H + K) &\leq \mathcal{B}_{\text{ph},L}(H) + \mathcal{B}_{\text{ph},L}(K), \\ \mathcal{B}_{\text{sin},L}(H + K) &\leq \mathcal{B}_{\text{sin},L}(H) + \mathcal{B}_{\text{sin},L}(K), \\ \mathcal{T}_{2,L}(H + K) &\leq \mathcal{T}_{2,L}(H) + \mathcal{T}_{2,L}(K).\end{aligned}\tag{S28}$$

For a nonzero real scalar λ , $\mathcal{K}_X(\lambda B) = \mathcal{K}_X(B)$ and $\Gamma_{\lambda B} = \Gamma_B$, hence all three atomic costs are unchanged, while $\|\lambda B\| = |\lambda|\|B\|$. Scaling decompositions in both directions gives

$$\begin{aligned}\mathcal{B}_{\text{ph},L}(\lambda H) &= |\lambda|\mathcal{B}_{\text{ph},L}(H), \\ \mathcal{B}_{\text{sin},L}(\lambda H) &= |\lambda|\mathcal{B}_{\text{sin},L}(H), \\ \mathcal{T}_{2,L}(\lambda H) &= |\lambda|\mathcal{T}_{2,L}(H).\end{aligned}\tag{S29}$$

The zero case is immediate. These functionals can vanish on nonzero operators that commute with the twist, so they are seminorms. They are independent of a chosen interaction presentation, but remain finite-volume quantities relative to the fixed lattice and on-site charge representation.

The infima also have a canonical envelope property. Let p be any seminorm on globally $U(1)$ -invariant Hermitian operators such that $p(B) \leq s_L(B)^2\|B\|$ for every labelled admissible supported term. For every admissible decomposition,

$$p(H) \leq \sum_a p(B_a) \leq \sum_a s_L(B_a)^2\|B_a\|.\tag{S30}$$

Taking the infimum proves $p(H) \leq \mathcal{B}_{\text{sin},L}(H)$. Conversely, the one-term decomposition shows that $\mathcal{B}_{\text{sin},L}$ is bounded by every displayed atomic cost. Thus it is the pointwise largest seminorm with those one-term bounds. The same argument with $s_{\text{ph},L}(B)^2\|B\|$ or $\tau_L(B)^2\|B\|$ gives the corresponding statements for $\mathcal{B}_{\text{ph},L}$ and $\mathcal{T}_{2,L}$.

These envelope seminorms also have exact polar representations. Let V_L be the finite-dimensional real vector space of globally $U(1)$ -invariant Hermitian operators, let $w(B, X)$ be any of $s_{\text{ph},L}(B)^2\|B\|$, $s_L(B)^2\|B\|$, or $\tau_L(B)^2\|B\|$, and let p_w denote the corresponding seminorm. Then

$$p_w(H) = \max_{\ell \in V_L^*} \{\ell(H) : |\ell(B)| \leq w(B, X) \text{ for every admissible labelled term } (B, X)\}.\tag{S31}$$

Indeed, every feasible ℓ bounds every decomposition term by term, hence $\ell(H) \leq p_w(H)$. Conversely, if $p_w(H) > 0$, the functional $f(aH) = ap_w(H)$ on $\mathbb{R}H$ is dominated by the sublinear functional p_w . The real Hahn–Banach theorem extends it to $\ell \in V_L^*$ with $\ell(K) \leq p_w(K)$ for all K . Applying this to $-K$ gives $|\ell(K)| \leq p_w(K)$, and the one-term decomposition gives $p_w(B) \leq w(B, X)$. Thus the extension is feasible and attains $\ell(H) = p_w(H)$. If $p_w(H) = 0$, the zero functional suffices. This also shows that a polar lower certificate need not assume that the primal decomposition infimum is attained.

Exact XY evaluation. Let $L \geq 3$, let each site carry spin one-half, and let

$$H_{\text{hop}} = \sum_j (t_j S_j^+ S_{j+1}^- + \bar{t}_j S_j^- S_{j+1}^+).\tag{S32}$$

Suppose Z is Hermitian and globally $U(1)$ invariant, $\mathcal{T}_{2,L}(Z) = 0$, and its restriction to the one-magnon sector has no nearest-neighbor off-diagonal matrix elements. Then

$$\boxed{\mathcal{T}_{2,L}(H_{\text{hop}} + Z) = \sum_j |t_j|}.\tag{S33}$$

To prove the lower bound, write $|j\rangle$ for the product state with one up spin at j and all other spins down. For an arbitrary admissible term B supported on X , form a graph G_B on X by joining $u \neq v$ whenever $\langle u|B|v\rangle \neq 0$. Charge conservation prevents one-magnon edges from entering or leaving X , and every graph edge supplies the transfer row $e_u - e_v$. Hence the weighted-charge commutant is constant on each nontrivial component C of G_B , and dropping its remaining constraints gives

$$\tau_L(B) \geq \sum_C d_L(C), \quad d_L(C) = \min_c \sum_{j \in C} \text{dist}_L(j, c).\tag{S34}$$

Let $m(C)$ be the number of nearest-neighbor cycle edges with both endpoints in C . If C is proper, $m(C) \leq |C| - 1$ and circular packing gives

$$d_L(C) \geq \left\lfloor \frac{|C|^2}{4} \right\rfloor \geq |C| - 1. \quad (\text{S35})$$

Indeed, the closest $|C|$ distinct lattice sites to a circular center have distances $0, 1, 1, \dots$ for odd $|C|$ and $\frac{1}{2}, \frac{1}{2}, \frac{3}{2}, \frac{3}{2}, \dots$ for even $|C|$; both sums equal $\lfloor |C|^2/4 \rfloor$. If C is the whole cycle, then $m(C) = L$ and $d_L(C) = \lfloor L^2/4 \rfloor$, whose square is at least L . Consequently

$$\sum_C m(C) \leq \sum_C d_L(C)^2 \leq \left(\sum_C d_L(C) \right)^2 \leq \tau_L(B)^2. \quad (\text{S36})$$

Since every matrix element is bounded by $\|B\|$,

$$\sum_j |\langle j|B|j+1 \rangle| \leq \tau_L(B)^2 \|B\|. \quad (\text{S37})$$

Set $\eta_j = t_j/|t_j|$ when $t_j \neq 0$ and $\eta_j = 0$ otherwise, and define the real linear functional

$$\ell_t(K) = \sum_j \text{Re}[\overline{\eta_j} \langle j|K|j+1 \rangle]. \quad (\text{S38})$$

Equation (S37) gives $|\ell_t(B)| \leq \tau_L(B)^2 \|B\|$ for every admissible term, so ℓ_t is feasible in Eq. (S31). The hypothesis on Z gives $\ell_t(Z) = 0$, while $\ell_t(H_{\text{hop}}) = \sum_j |t_j|$. The polar formula therefore supplies the lower certificate $\mathcal{T}_{2,L}(H_{\text{hop}} + Z) \geq \sum_j |t_j|$. Subadditivity and the zero-seminorm hypothesis give equality because each hopping bond has transport radius one and norm $|t_j|$. This proves Eq. (S33).

The finite-angle seminorm can also be evaluated exactly throughout the model range $L \geq 6$. Let A be the one-magnon restriction of an admissible term B , and let F be the set of cycle edges j for which $\langle j|B|j+1 \rangle \neq 0$. If F is nonempty, choose a Hermitian signed edge matrix R_B whose entry on each oriented edge is the conjugate phase of $\langle j|B|j+1 \rangle$. Compression to the one-magnon sector gives $\|A\| \leq \|B\|$, while the two orientations of each Hermitian edge give $\text{Tr}(R_B A) = 2 \sum_j |\langle j|B|j+1 \rangle|$. Schatten 1- ∞ duality therefore gives

$$W_B := \sum_j |\langle j|B|j+1 \rangle| = \frac{1}{2} \text{Tr}(R_B A) \leq g(F) \|B\|, \quad g(F) := \frac{1}{2} \|R_B\|_*. \quad (\text{S39})$$

If F is not the full cycle, write its path-run edge lengths as $r_1, \dots, r_c$. Gauging away the phases on each path gives

$$g(F) = \sum_{i=1}^c \sum_{k=1}^{r_i+1} \left| \cos \frac{k\pi}{r_i+2} \right|, \quad D(F) = \sum_{i=1}^c \left\lfloor \frac{(r_i+1)^2}{4} \right\rfloor. \quad (\text{S40})$$

Indeed, a run of r_i edges is unitarily equivalent to the adjacency matrix of a path on $r_i + 1$ vertices. Its eigenvalues are $2 \cos[k\pi/(r_i + 2)]$, $k = 1, \dots, r_i + 1$; half of their absolute sum is the corresponding summand in Eq. (S40). This also makes explicit why the aligned edge phases do not affect the nuclear norm. For the full cycle put $D(F) = \lfloor L^2/4 \rfloor$. Distinct runs have disjoint vertex sets, and merging them by additional edges cannot reduce the sum of circular median dispersions. Thus Eq. (S34) gives $\tau_L(B) \geq D(F)$. Moreover, subadditivity of the nuclear norm over single-edge matrices gives $g(F) \leq |F| \leq D(F)$.

Put $x = \pi/L$. If $D(F) < L/2$, then $D(F) \sin^2 x \leq \sin^2(D(F)x)$: the cases $D = 1, 2$ are direct, while for $D \geq 3$ the sine chord gives $\sin(Dx) \geq (2D/\pi) \sin x \geq \sqrt{D} \sin x$. If $D(F) \geq L/2$, the finite-angle cost is clipped to one. For $L \geq 7$, the Frobenius bound, with $m = |F|$ and at most $n \leq L$ incident vertices, gives $g(F) \leq \sqrt{mn/2} \leq L/\sqrt{2}$ and hence $g(F) \sin^2(\pi/L) \leq \pi^2/(\sqrt{2}L) < 1$. For $L = 6$, the cases $m \leq 4$ follow from $g(F) \leq m$. If $m = 5$, the six-vertex path has $g(F) = \csc(\pi/14) - 1 < 4$; if $m = 6$, pairing antipodal cycle eigenvalues gives $g(F) = 2 \sum_{k=0}^2 |\cos(\theta + k\pi/3)| \leq 4$. Since $\sin^2(\pi/6) = 1/4$, the endpoint also follows. Together with Eq. (S39), both regimes yield

$$\sin^2 \frac{\pi}{L} \sum_j |\langle j|B|j+1 \rangle| \leq s_L(B)^2 \|B\|. \quad (\text{S41})$$

Thus $\ell_{\sin} = \sin^2(\pi/L)\ell_t$ is feasible in the finite-angle polar formula (S31). It has value $\sin^2(\pi/L)\sum_j |t_j|$ on $H_{\text{hop}} + Z$. Since $\mathcal{B}_{\sin,L}(Z) \leq (2\pi S/L)^2 \mathcal{T}_{2,L}(Z) = 0$, the hopping-bond upper certificate matches it. Hence

$$\boxed{\mathcal{B}_{\sin,L}(H_{\text{hop}} + Z) = \sin^2 \frac{\pi}{L} \sum_j |t_j|, \quad L \geq 6.} \quad (\text{S42})$$

For the uniform XY ring, $t_j = J/2$, so $\mathcal{T}_{2,L}(H_L^{\text{XY}}) = L|J|/2$. By contrast, using the same Hamiltonian as one full-ring term gives $\tau_L(H_L^{\text{XY}}) = \lfloor L^2/4 \rfloor$ and costs at least $|J|\lfloor L^2/4 \rfloor^2$. Thus the bond decomposition attains the infimum and removes a genuine presentation-dependent inflation. More generally, let $p \mid L$, let $t_{j+p} = t_j$, and let Z obey the hypotheses above and commute with T_L^p . Equations (S3), (S25), and (S42) give, whenever $pN \notin L\mathbb{Z}$,

$$\text{gap}_N(H_{\text{hop}} + Z) \leq 2 \sin^2 \frac{\pi}{L} \sum_j |t_j| = \frac{2L}{p} \sin^2 \frac{\pi}{L} \sum_{a=0}^{p-1} |t_a|. \quad (\text{S43})$$

For real p -periodic exchange and DM couplings,

$$J_j(S_j^x S_{j+1}^x + S_j^y S_{j+1}^y) + D_j(S_j^x S_{j+1}^y - S_j^y S_{j+1}^x), \quad (\text{S44})$$

one has $t_j = (J_j + iD_j)/2$, which proves Eq. (14). At $p = 2$ and half filling, $pN/L = 1$, so this corollary correctly makes no claim about the dimerization gap.

TRANSFER MATRICES AND CERTIFICATES

Let $|a\rangle$ be the product S^z basis and write $M(c) = \sum_j c_j S_j^z$. Its commutator has matrix elements

$$\langle a|[B, M(c)]|b\rangle = \langle a|B|b\rangle \sum_j c_j [m_j(b) - m_j(a)]. \quad (\text{S45})$$

Thus $[B, M(c)] = 0$ exactly when c is orthogonal to every transfer vector associated with a nonzero matrix element. If D_B has these vectors as rows and $x = (j)_{j \in X}$, then

$$\begin{aligned} \mathcal{K}_X(B) &= \ker_{\mathbb{R}} D_B, \\ \tau_L(B) &= \inf_{n \in \mathbb{Z}^X} \text{dist}_1(x + Ln, \ker D_B). \end{aligned} \quad (\text{S46})$$

For completeness, let K be a linear subspace of a finite-dimensional real vector space equipped with ℓ^1 . The dual of the quotient by K is the annihilator $K^\perp$ equipped with the restricted ℓ^∞ norm. Finite-dimensional norm duality gives

$$\text{dist}_1(v, K) = \max_{y \in K^\perp, \|y\|_\infty \leq 1} y \cdot v. \quad (\text{S47})$$

Since $(\ker D_B)^\perp = \text{ran } D_B^T$, this proves the corresponding transfer-matrix dual certificate. Exact matrix zeros therefore furnish a rigorous certificate of the commutant and its quotient cost.

Exact rank-one formula. Suppose the nonzero integer rows of D_B span a one-dimensional real space, and let $d \in \mathbb{Z}^X$ be its primitive generator, unique up to sign, with $\gcd_j |d_j| = 1$. Then

$$\boxed{\tau_L(B) = \frac{\text{dist}(d \cdot x, L\mathbb{Z})}{\|d\|_\infty}.} \quad (\text{S48})$$

Indeed, $\ker D_B = \ker d$, and Eq. (S47) gives, for every fixed integer lift $v = x + Ln$,

$$\text{dist}_1(v, \ker d) = \frac{|d \cdot v|}{\|d\|_\infty}. \quad (\text{S49})$$

Primitivity and Bézout's identity imply $\{d \cdot n : n \in \mathbb{Z}^X\} = \mathbb{Z}$; minimizing over n proves Eq. (S48). Shifting a representative by a multiple of L changes $d \cdot x$ by a multiple of L , and changing the sign of d does not change the result.

For a spin-one-half fixed-transfer monomial, the transfer vector $r \in \{-1, 0, 1\}^X$ is primitive and the Hermitian conjugate contributes $-r$. Hence

$$\tau_L(B) = \text{dist} \left(\sum_{j \in X} r_j j, L\mathbb{Z} \right). \quad (\text{S50})$$

It is zero precisely when the wrapped polarization displacement is a multiple of the circumference. This includes ordinary dipole-neutral processes and processes that wrap the chosen seam. In higher-spin representations an actual transfer may be a nonprimitive multiple of d ; its finite twist phase can then cancel even when Eq. (S48) is positive. The formula concerns the real transfer row space defining τ_L , not the integer row lattice.

Atomic transport radius is not closed under sums, which is why the Hamiltonian decomposition infimum is essential. For example, take $L = 6$, $x = (0, 1, 3, 4)$, and two Hermitian spin-one-half fixed-transfer terms with

$$d_1 = (-1, -1, 1, 1), \quad d_2 = (-1, 1, 1, -1). \quad (\text{S51})$$

Their moments are 6 and 0, so each term has zero radius by Eq. (S48). The transfer kernel of their sum is $\{(a, b, a, b) : a, b \in \mathbb{R}\}$. For every integer lift $v = x + 6n$,

$$\text{dist}_1(v, \ker D) = |v_0 - v_2| + |v_1 - v_3| \geq 3 + 3 = 6, \quad (\text{S52})$$

and equality is attained, so the combined atomic radius is 6. Nevertheless, retaining the two summands as separate admissible terms proves that all three Hamiltonian seminorms of their sum vanish.

The transfer matrix and dual witness certify the cost of a specified term and lift. Efficient optimization over all Hamiltonian decompositions is a separate algorithmic problem. Any explicit decomposition gives a rigorous upper bound on the seminorm and therefore on the sector gap.

Suppose the transfer row space is generated by $e_i - e_j$ for the edges of a graph. A vector lies in the kernel exactly when it is constant on each graph component. Both the integer lifts and constants separate componentwise, and minimizing their ℓ^1 costs gives the sum of circular median dispersions.

FINITE-RANGE GEOMETRY

For $X \subset \Lambda_L$, define its circular ℓ^1 dispersion by

$$d_L(X) = \min_{c \in \mathbb{R}/L\mathbb{Z}} \sum_{j \in X} \text{dist}_L(j, c), \quad (\text{S53})$$

where dist_L is the shortest distance on the cycle.

For reference, suppose a supported decomposition satisfies

$$\sup_x \sum_{X \ni x} \|\Phi_X\| \leq J \quad (\text{S54})$$

and every support has cycle diameter at most R .

Let X have cycle diameter at most R . If $L > 2R$, fix an anchor in X and choose the unique principal displacement of every other site in $[-R, R] \cap \mathbb{Z}$. Distinct sites have distinct displacements. The worst absolute sum uses every nonzero integer in that interval, hence

$$d_L(X) \leq R(R+1). \quad (\text{S55})$$

If $L > 3R$, the support can be lifted to an integer interval of diameter at most R . Among at most one site at each of the $R+1$ integer positions, the sum of absolute deviations is maximized by occupying every position. A median minimizes that sum, and direct summation gives

$$\max_{Y \subset \{0, \dots, R\}} \min_{c \in \mathbb{R}} \sum_{j \in Y} |j - c| = \left\lfloor \frac{(R+1)^2}{4} \right\rfloor. \quad (\text{S56})$$

The full consecutive set attains equality. Applying the constant-vector choice in Eq. (4) proves the geometric estimates used in the following finite-range bounds.

Finally, summing Eq. (S54) over sites gives

$$\sum_X |X| \|\Phi_X\| \leq LJ. \quad (\text{S57})$$

Every nonempty term has $|X| \geq 1$, and conditional expectation contracts norm, so $\sum_X \|B_X\| \leq LJ$.

Together with Eq. (S2), the two geometric estimates give, for every eigenvalue $pN \notin L\mathbb{Z}$,

$$\text{gap}_N(H_L) \leq \frac{8\pi^2 S^2 R^2 (R+1)^2 J}{L}, \quad L > 2R, \quad (\text{S58})$$

$$\text{gap}_N(H_L) \leq \frac{8\pi^2 S^2 D_R^2 J}{L}, \quad L > 3R, \quad (\text{S59})$$

where $D_R = \lfloor (R+1)^2/4 \rfloor$.

LONG-RANGE TRANSPORT CLASSES

This section specializes to one-site translation, $p = 1$. All sector bounds below hold for every eigenvalue $N \notin L\mathbb{Z}$.

For one translation orbit of two-site exchange terms at each cycle distance r , assume $\|B_{j,r}\| \leq C(1+r)^{-\alpha}$. Since $\tau_L(B_{j,r}) \leq r$, Eq. (S3) implies, up to the convention at the even-ring antipode,

$$\text{gap}_N(H_L) \leq \frac{8\pi^2 S^2 C}{L} \sum_{r=1}^{\lfloor L/2 \rfloor} \frac{r^2}{(1+r)^\alpha}. \quad (\text{S60})$$

This is $O(L^{-1})$ for $\alpha > 3$, $O(L^{-1} \log L)$ at $\alpha = 3$, and $O(L^{2-\alpha})$ for $2 < \alpha < 3$; in particular, the bound vanishes for $\alpha > 2$.

For a second class, assume directly that $\tau_L(B_{j,r}) \leq K$ uniformly in j, r, L ; products of two separately $U(1)$ -neutral clusters with uniformly bounded internal dispersion provide the standard example. The same argument gives

$$\text{gap}_N(H_L) \leq \frac{8\pi^2 S^2 C K^2}{L} \sum_{r=1}^{\lfloor L/2 \rfloor} \frac{1}{(1+r)^\alpha}. \quad (\text{S61})$$

It is $O(L^{-1})$ for $\alpha > 1$, $O(L^{-1} \log L)$ at $\alpha = 1$, and $O(L^{-\alpha})$ for $0 < \alpha < 1$. These are the Type-I and Type-II one-dimensional thresholds of Ref. [7], now expressed through the same transport functional.

In particular, for any sequence of rings and any chosen fractional-sector sequence $N_L \notin L\mathbb{Z}$,

$$\mathcal{T}_{2,L}(H_L) = o(L^2) \implies \text{gap}_{N_L}(H_L) \longrightarrow 0. \quad (\text{S62})$$

DIRECTIONAL CYLINDERS AND DIMENSIONAL CROSSOVER

Let

$$\Lambda_L = \mathbb{Z}/L\mathbb{Z} \times F_L, \quad W_L = |F_L|, \quad (\text{S63})$$

where the finite fiber F_L may depend on L . At every (x, a) let $q_{x,a}$ be a finite-dimensional self-adjoint charge with integer spectrum, and define

$$Q_L = \sum_{x,a} q_{x,a}, \quad U_L^\parallel = \exp\left(\frac{2\pi i}{L} \sum_{x=0}^{L-1} \sum_{a \in F_L} x q_{x,a}\right). \quad (\text{S64})$$

The translation T_L advances x and preserves a . Integer charge then gives the exact relation

$$T_L^p U_L^\parallel T_L^{-p} = U_L^\parallel e^{-2\pi i p Q_L / L}. \quad (\text{S65})$$

We assume $p \mid L$, $[H_L, Q_L] = 0$, and $[H_L, T_L^p] = 0$.

For a Hermitian term B supported on $X \subset \Lambda_L$ with $[B, Q_X] = 0$, repeat the joint-charge projector-graph construction, now assigning the phase

$$\theta_\lambda^\parallel = \frac{2\pi}{L} \sum_{(x,a) \in X} x \lambda_{x,a} \pmod{2\pi}. \quad (\text{S66})$$

Let $\eta_{\parallel,L}(B)$ be the largest shortest-containing-arc radius of its graph components and put

$$s_{\text{ph},L}^\parallel(B) = \sin(\min\{\eta_{\parallel,L}(B), \pi/2\}). \quad (\text{S67})$$

The corresponding Hamiltonian seminorm is

$$\mathcal{B}_{\text{ph},L}^\parallel(H) = \inf_{H=\sum_j B_j} \sum_j (s_{\text{ph},L}^\parallel(B_j))^2 \|B_j\|, \quad (\text{S68})$$

over finite supported Hermitian decompositions with termwise charge conservation.

Directional-cylinder theorem. For every charge eigenvalue N satisfying $pN \notin L\mathbb{Z}$,

$$\boxed{\text{gap}_N(H_L) \leq 2\mathcal{B}_{\text{ph},L}^\parallel(H_L)}. \quad (\text{S69})$$

To prove this, first apply the global $U(1)$ conditional expectation to any supported Hermitian decomposition. It contracts every norm and produces $[B_X, Q_X] = 0$. This expectation commutes with conjugation by the local longitudinal twist because both are generated by the mutually commuting on-site charges; locally twist-invariant terms therefore remain invariant. The exact charge-logarithm theorem and paired-conjugation inequality give

$$\begin{aligned} & \left\| \frac{(U_L^\parallel)^{-1} B_X U_L^\parallel + U_L^\parallel B_X (U_L^\parallel)^{-1}}{2} - B_X \right\| \\ & \leq 2(s_{\text{ph},L}^\parallel(B_X))^2 \|B_X\|. \end{aligned} \quad (\text{S70})$$

Equation (S65) makes each of $(U_L^\parallel)^{\pm 1} \Psi_{L,N}$ orthogonal to a unique sector ground state $\Psi_{L,N}$, although the two twisted states need not be mutually orthogonal. Averaging their excitation energies, applying sector min-max, and taking the decomposition infimum proves Eq. (S69). A degenerate sector ground energy makes the sector gap zero and is automatic. The same proof applies to any selected periodic direction of a finite product torus; it does not combine different twists into a higher-dimensional flux-insertion theorem.

Exact transverse invisibility. If B is supported in one slice $x = x_0$, its local longitudinal twist is $e^{2\pi i x_0 Q_X / L}$ and hence commutes with B . Therefore

$$\eta_{\parallel,L}(B) = 0. \quad (\text{S71})$$

Arbitrary charge-conserving transverse hopping, intra-slice many-body interactions, and intra-slice disorder have zero longitudinal cost, independently of their strength, order, and transverse range. Product-charge diagonal interactions of arbitrary longitudinal range also have zero cost.

Selective hopping on a graph fiber. Let every site have integer levels $0, 1, \dots, M$, $M \geq 1$. Choose one oriented representative e of each physical Hermitian hopping bond from (x, a) to $(x + r_e, b)$ and write

$$H_L = \sum_e (t_e |1, 0\rangle \langle 0, 1|_e + \bar{t}_e |0, 1\rangle \langle 1, 0|_e) + D_L. \quad (\text{S72})$$

Assume the edge collection and amplitudes are p -periodic in x , $[D_L, Q_L] = [D_L, T_L^p] = 0$, and D_L has a supported decomposition into charge-conserving terms commuting with their local longitudinal twists. Writing

$$d_L(r) = \min_{n \in \mathbb{Z}} |r + nL|, \quad (\text{S73})$$

the edge term has norm $|t_e|$ and phase radius $\pi d_L(r_e)/L$. Equation (S69) yields

$$\boxed{\text{gap}_N(H_L) \leq 2 \sum_e |t_e| \sin^2 \left(\frac{\pi d_L(r_e)}{L} \right), \quad pN \notin L\mathbb{Z}.} \quad (\text{S74})$$

This is independent of transverse separation, unused charge levels, and all admitted terms in D_L . Edges with $r_e = 0$ contribute exactly zero.

For the fiber-preserving nearest-neighbor subclass $((x, a), (x + 1, a))$, write $t_{x,a} = t_{c,a}$ with $c = x \bmod p$. Then

$$\text{gap}_N(H_L) \leq \frac{2L}{p} \sin^2 \frac{\pi}{L} \sum_{a \in F_L} \sum_{c=0}^{p-1} |t_{c,a}|. \quad (\text{S75})$$

For uniform $t_{c,a} = t$,

$$\text{gap}_N(H_L) \leq 2LW_L |t| \sin^2 \frac{\pi}{L} \leq \frac{2\pi^2 W_L |t|}{L}. \quad (\text{S76})$$

Consequently the bound vanishes when $\sup_{L,c,a} |t_{c,a}| < \infty$ and $W_L = o(L)$, provided the fractional-sector gate persists. More generally it is enough that the cell sum in Eq. (S75) is $O(pW_L)$. At isotropic aspect ratio $W_L \asymp L$, this elementary uniform-twist estimate is only $O(1)$ and makes no two-dimensional inapplicability claim. When $W_L = 1$, Eq. (S74) recovers the former multilevel-chain bound; its independence of M removes the spurious M^2 factor generated by the charge-width certificate in the small-angle quadratic relaxation.

Power-law moving profile. For $1 \leq r \leq n := \lfloor L/2 \rfloor$, define the actual longitudinal ℓ^1 hopping profile

$$A_L(r) = \sum_{e: d_L(r_e)=r} |t_e|. \quad (\text{S77})$$

Grouping Eq. (S74) by r gives the exact profile estimate

$$\text{gap}_N(H_L) \leq 2 \sum_{r=1}^n A_L(r) \sin^2 \frac{\pi r}{L}. \quad (\text{S78})$$

Suppose that, for constants $C \geq 0$, $\alpha > 2$, and a nonnegative channel mass K_L ,

$$A_L(r) \leq CLK_L r^{-\alpha}. \quad (\text{S79})$$

Then $\sin x \leq x$ yields

$$\begin{aligned} \text{gap}_N(H_L) &\leq 2CLK_L \sum_{r=1}^n r^{-\alpha} \sin^2 \frac{\pi r}{L} \\ &\leq \frac{2\pi^2 CK_L}{L} \sum_{r=1}^n r^{2-\alpha}. \end{aligned} \quad (\text{S80})$$

The elementary moment bounds are

$$\sum_{r=1}^n r^{2-\alpha} \leq \begin{cases} \zeta(\alpha - 2), & \alpha > 3, \\ 1 + \log n, & \alpha = 3, \\ 1 + \frac{n^{3-\alpha} - 1}{3 - \alpha}, & 2 < \alpha < 3. \end{cases} \quad (\text{S81})$$

Consequently, if $K_L = O(L^\beta)$, the three scalings are respectively $O(L^{\beta-1})$, $O(L^{\beta-1} \log L)$, and $O(L^{\beta+2-\alpha})$. The bound therefore vanishes in the strict region

$$\beta < \min\{1, \alpha - 2\}, \quad (\text{S82})$$

with $K_L = o(L/\log L)$ at the dipolar boundary $\alpha = 3$. The assertion $K_L = O(W_L)$ is an assumption on the actual inter-slice moving-channel mass. It follows, for example, when each distance and longitudinal cell has only $O(W_L)$ moving bonds with uniformly bounded power-law prefactors; bounded degree of the fiber graph alone is not sufficient. Dense inter-slice graphs may instead have a larger channel mass, whereas arbitrary purely transverse charge-conserving dynamics remains exactly zero cost.

For an even-length odd-leg $\text{SU}(2)$ -invariant ladder, replace the longitudinal part of Eq. (S94) by exchanges $J_{a,r}$ at shortest distance r and assume

$$\sum_a |J_{a,r}| \leq CW_L r^{-\alpha}. \quad (\text{S83})$$

There are at most L physical bonds for each (a, r) , with a smaller antipodal orbit when L is even. The charge-moving part of a Heisenberg bond has norm $|J_{a,r}|/2$, so the factor two in Eq. (S74) cancels and gives

$$\text{gap}_{\text{full}}(H_L) \leq L \sum_{r,a} |J_{a,r}| \sin^2 \frac{\pi r}{L} \leq \frac{\pi^2 C W_L}{L} \sum_{r=1}^n r^{2-\alpha}. \quad (\text{S84})$$

The unique-ground-state singlet argument below supplies the full-gap gate; degenerate ground energy again makes the bound automatic.

Dense Euclidean strip. Specialize to one-site translation, $p = 1$. Now take $F_L = \mathbb{Z}/W_L\mathbb{Z}$, write $W = W_L \leq n$, and allow every pair of transverse channels to be coupled. For one oriented representative of each physical Hermitian longitudinal bond let $r_e = d_L(\Delta x_e) \geq 1$ and $s_e = d_W(\Delta y_e) \geq 0$, and assume

$$|t_e| \leq J(r_e^2 + s_e^2)^{-\alpha/2}, \quad \alpha > 2. \quad (\text{S85})$$

For $1 \leq s \leq \lfloor W/2 \rfloor$, set

$$m_W(s) = \begin{cases} 1, & W \text{ even and } s = W/2, \\ 2, & \text{otherwise.} \end{cases} \quad (\text{S86})$$

Thus $1 + \sum_s m_W(s) = W$, including the transverse antipode once. When L is even, $r = L/2$ is represented by the half longitudinal translation orbit; translation covariance then requires the corresponding channel matrix to be Hermitian, with self-reversing simultaneous-antipode entries real, so every physical bond is counted exactly once. Counting longitudinal translates and transverse sources gives

$$A_{L,W}(r) \leq J L W \left[r^{-\alpha} + \sum_{s=1}^{\lfloor W/2 \rfloor} m_W(s) (r^2 + s^2)^{-\alpha/2} \right]. \quad (\text{S87})$$

Define

$$\kappa_\alpha = 1 + \frac{\sqrt{\pi} \Gamma((\alpha - 1)/2)}{\Gamma(\alpha/2)}. \quad (\text{S88})$$

The decreasing transverse summand and its integral over the half-line give

$$\begin{aligned} r^{-\alpha} + \sum_s m_W(s) (r^2 + s^2)^{-\alpha/2} &\leq \kappa_\alpha r^{1-\alpha}, \\ r^{-\alpha} + \sum_s m_W(s) (r^2 + s^2)^{-\alpha/2} &\leq W r^{-\alpha}. \end{aligned} \quad (\text{S89})$$

Using the first estimate for $r \leq W$ and the second for $r > W$ in Eq. (S78), followed by $\sin x \leq x$, yields

$$\text{gap}_N(H_L) \leq \frac{2\pi^2 J W}{L} \left[\kappa_\alpha \sum_{r=1}^W r^{3-\alpha} + W \sum_{r=W+1}^n r^{2-\alpha} \right]. \quad (\text{S90})$$

Elementary integral comparisons now give the explicit branches

$$\text{gap}_N(H_L) \leq \begin{cases} \frac{2\pi^2 J W}{L} \left[\kappa_\alpha \zeta(\alpha - 3) + \frac{1}{\alpha - 3} \right], & \alpha > 4, \\ \frac{2\pi^2 J W}{L} [(1 + \pi/2)(1 + \log W) + 1], & \alpha = 4, \\ \frac{2\pi^2 J W^{5-\alpha}}{L} \left[\frac{\kappa_\alpha}{4 - \alpha} + \frac{1}{\alpha - 3} \right], & 3 < \alpha < 4, \\ \frac{2\pi^2 J W^2}{L} \left[3 + \log \frac{n}{W} \right], & \alpha = 3, \\ \frac{2\pi^2 J W^2}{L} n^{3-\alpha} \left[\kappa_\alpha + \frac{1}{3 - \alpha} \right], & 2 < \alpha < 3. \end{cases} \quad (\text{S91})$$

For the dipolar endpoint, the constant and logarithm can be read off without hiding the discrete–continuous comparison:

$$\kappa_3 = 1 + \frac{\sqrt{\pi}\Gamma(1)}{\Gamma(3/2)} = 3, \quad \sum_{r=W+1}^n \frac{1}{r} \leq \int_W^n \frac{dx}{x} = \log \frac{n}{W}. \quad (\text{S92})$$

Thus the $r \leq W$ part of Eq. (S90) contributes $3W$, and the $r > W$ part contributes at most $W \log(n/W)$ inside the brackets, yielding the constant 3 in the $\alpha = 3$ branch after restoring the common prefactor. For $W = O(L^\beta)$ the corresponding strict polynomial-width thresholds are

$$\beta < \begin{cases} (\alpha - 2)/2, & 2 < \alpha \leq 3, \\ 1/(5 - \alpha), & 3 \leq \alpha < 4, \\ 1, & \alpha \geq 4, \end{cases} \quad (\text{S93})$$

with logarithmic refinements at $\alpha = 3, 4$. In particular, fully connected dipolar exchange obeys Eq. (17) and vanishes when $W^2[1 + \log(L/W)] = o(L)$, while a charge-moving van der Waals tail is $O(W/L)$. This is a sufficient anisotropic aspect-ratio region, not a converse or an isotropic two-dimensional theorem. Dipolar spin-exchange XY dynamics in polar-molecule arrays provide the relevant U(1)-conserving mechanism [15–17], but we do not assert that those experiments realize the precise periodic strip geometry, filling, or spectral hypotheses used here. By contrast, a standard coherently driven Rydberg Ising simulator [21] does not conserve Rydberg occupation and is not covered merely because its density interaction is van der Waals.

Odd-leg spin-one-half ladder. Let L be even and W_L odd, and consider

$$H_L = \sum_{x=0}^{L-1} \sum_{a=1}^{W_L} J_a \mathbf{S}_{x,a} \cdot \mathbf{S}_{x+1,a} + \sum_{x=0}^{L-1} K_x + Z_L. \quad (\text{S94})$$

Here J_a are real, the K_x are translates of an arbitrary SU(2)-invariant Hermitian operator supported within one rung, and Z_L is translation and SU(2) invariant with a supported decomposition into terms commuting with their local longitudinal twists. No norm or interaction-order bound is placed on K_x or the admitted terms of Z_L .

Use $q_{x,a} = S_{x,a}^z + 1/2$. Every rung term has zero longitudinal cost. The charge-moving part of a longitudinal Heisenberg bond has norm $|J_a|/2$ and phase radius π/L . If the ground state is degenerate, the full gap is zero. Otherwise SU(2) invariance makes it a singlet with $N = LW_L/2 \notin L\mathbb{Z}$. Hence

$$\text{gap}_{\text{full}}(H_L) \leq L \sin^2 \frac{\pi}{L} \sum_{a=1}^{W_L} |J_a| \leq \frac{\pi^2}{L} \sum_{a=1}^{W_L} |J_a|. \quad (\text{S95})$$

For uniform longitudinal exchange this is $\pi^2 W_L |J_{\parallel}|/L$ and vanishes for bounded $J_{\parallel}$ whenever $W_L = o(L)$. Fixed-width ladder LSM statements are known; the formulation used here records the direction-resolved finite-volume cost, the explicit growing-width crossover, and exact independence from arbitrary admitted rung dynamics.

DIAGONAL-TAIL AND POLARIZATION-NEUTRAL HOPPING MODEL

We derive Eq. (12) with all finite-size factors explicit. For spin one-half, write the transverse bond as

$$B_j = J_{\perp} (S_j^x S_{j+1}^x + S_j^y S_{j+1}^y) = \frac{J_{\perp}}{2} (S_j^+ S_{j+1}^- + S_j^- S_{j+1}^+). \quad (\text{S96})$$

On the subspace spanned by $\{|\uparrow\downarrow\rangle, |\downarrow\uparrow\rangle\}$, it is the matrix with off-diagonal entries $J_{\perp}/2$ and it vanishes on the two remaining product states. Thus

$$\|B_j\| = \frac{|J_{\perp}|}{2}. \quad (\text{S97})$$

For $J_{\perp} \neq 0$, its transfer graph is the single edge $(j, j+1)$, so $\tau_L(B_j) = 1$; for $J_{\perp} = 0$, $\tau_L(B_j) = 0$. In all cases $\tau_L(B_j) \leq 1$.

Every operator diagonal in the product S^z basis commutes with $\sum_j c_j S_j^z$ for every real vector c . Hence $\mathcal{K}_{\Lambda_L}(D_L) = \mathbb{R}^{\Lambda_L}$ and $\tau_L(D_L) = 0$ even when D_L is retained as one full-ring term.

Write $Z_{\text{pol}} = \sum_{\alpha} B_{\alpha}$, retaining each Hermitian fixed-transfer spin-one-half monomial as a separate admissible term. If its transfer vector $r^{(\alpha)}$ satisfies

$$\sum_j r_j^{(\alpha)} j \in L\mathbb{Z}, \quad (\text{S98})$$

then Eq. (S50) gives $\tau_L(B_{\alpha}) = 0$. A nontrivial zero-cost spin-one-half transfer cannot contain only one raising and one lowering operator: that would be a transfer between two distinct ring sites and would have nonzero cycle distance. Consequently every B_{α} has at least two raising and two lowering operators and annihilates the one-magnon sector. The displayed decomposition therefore proves

$$\mathcal{T}_{2,L}(Z_{\text{pol}}) = \mathcal{B}_{\text{sin},L}(Z_{\text{pol}}) = \mathcal{B}_{\text{ph},L}(Z_{\text{pol}}) = 0, \quad (\text{S99})$$

even though $\tau_L(Z_{\text{pol}})$ need not vanish.

For the counterflow term in Eq. (11), use the integer lifts

$$(\tilde{j}_1, \tilde{j}_2, \tilde{j}_3, \tilde{j}_4) = (j, j+1, j+r, j+r+1). \quad (\text{S100})$$

They are valid even when either bond crosses the chosen seam. The first monomial has transfer vector $d = (1, -1, -1, 1)$ and the conjugate monomial has $-d$, while

$$d \cdot (\tilde{j}_1, \tilde{j}_2, \tilde{j}_3, \tilde{j}_4) = 0. \quad (\text{S101})$$

Thus $C_{j,r}$ commutes with the weighted charge whose coefficients are these lifts. Choosing the same lifts and coefficient vector in Eq. (4) gives $\tau_L(C_{j,r}) = 0$ exactly. For $K_{r,L} \neq 0$ the individual term has nonzero off-diagonal product-basis matrix elements. When L is even and $r = L/2$, translation by r exchanges the two monomials, so the orbit sum depends only on $\text{Re } K_{r,L}$; at least one noncancelling nonzero orbit produces an actual off-diagonal interaction.

The diagonal term has no one-magnon hopping, and the same holds for Z_{pol} . With $Z = D_L + Z_{\text{pol}}$, Eq. (S99) and subadditivity give $\mathcal{T}_{2,L}(Z) = 0$. Equation (S33) therefore upgrades the physical decomposition bound to the exact identity

$$\mathcal{T}_{2,L}(H_L) = \frac{L|J_{\perp}|}{2}. \quad (\text{S102})$$

The first inequality in Eq. (S3), evaluated on the physical bond decomposition, already gives Eq. (12). For completeness, it can also be read directly from the exact twist response. Put $\theta = 2\pi/L$ and write $H_L = H_{\text{hop}} + Z$, where $Z = D_L + Z_{\text{pol}}$. Diagonality and Eq. (S98) make the uniform-twist phase of every displayed constituent of Z cancel exactly, while the two hopping directions acquire opposite phases. Therefore

$$\frac{U_L^{-1} H_{\text{hop}} U_L + U_L H_{\text{hop}} U_L^{-1}}{2} = \cos \theta H_{\text{hop}}, \quad [Z, U_L] = 0. \quad (\text{S103})$$

If a fractional-sector ground state Ψ is unique in that sector, let $e_{\pm}$ be the excitation energies of $U_L^{\pm 1} \Psi$. They are nonnegative and their trial states are orthogonal to Ψ by the translation phase derived above. Equation (S103) gives

$$\begin{aligned} 0 \leq \frac{e_+ + e_-}{2} &= (\cos \theta - 1) \langle \Psi, H_{\text{hop}} \Psi \rangle \\ &\leq (1 - \cos \theta) \|H_{\text{hop}}\| \\ &\leq \frac{L|J_{\perp}|}{2} (1 - \cos \theta). \end{aligned} \quad (\text{S104})$$

One twist sign lies below the average. Sector min-max, together with the automatic zero gap in the degenerate case, proves the sharper finite-ring bound

$$\text{gap}_N(H_L) \leq L|J_{\perp}| \sin^2 \frac{\pi}{L} \leq \frac{\pi^2 |J_{\perp}|}{L}, \quad N \notin L\mathbb{Z}. \quad (\text{S105})$$

The first inequality is strictly sharper than the quadratic upper bound whenever $J_{\perp} \neq 0$. Corollary C gives the corresponding full-gap bound when its global spectral condition holds.

For comparison, a diameter estimate assigns $r^2 \|V_r S_j^z S_{j+r}^z\| = r^2 |V_r|/4$ to each of the L terms at distance r . Multiplication by the spin-half prefactor $2\pi^2/L^2$ gives the extra diameter contribution

$$\frac{\pi^2}{2L} \sum_{r=1}^{\lfloor (L-1)/2 \rfloor} r^2 |V_r|. \quad (\text{S106})$$

For $V_r = V/r^\alpha$, this is exactly Eq. (18). The restriction $r \leq \lfloor (L-1)/2 \rfloor$ makes r the unambiguous shortest cycle distance; the general diagonal D_L may also contain the even-ring antipodal term.